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Laminar flow and pressure drop of complex fluids in a Sulzer SMX+™ static mixer

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ABSTRACT

Static mixers are an effective model system for investigating the effect of fluid rheology on pressure losses in laminar flow. Viscoelastic pressure drop enhancement in real process fluids was investigated using experimental methods and computer simulations of a SMX+™ mixer. The results were compared against classical friction factor correlations for non-elastic fluids and literature correlation for elastic fluids (developed using aqueous polymers). CFD predictions of pressure losses for both inelastic and viscoelastic process fluids were consistent with the measurements. The literature correlation for viscoelastic fluids progressively overestimated the friction factor with increasing Reynolds number. The CFD results were analysed to discern the impact of the flow field on laminar mixing.

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1. Introduction

Static mixers have been used for continuous operations in the chemical processing industry over the last 50 years. They offer comparative advantages over dynamic mixers as they are able to process fluids over a wide range of viscosities with minimal space and maintenance requirements (Myers et al., 1997; Bayer et al., 2003). They are particularly suited for the inline blending of high viscosity non-Newtonian fluids in the laminar flow regime (Jegatheeswaran et al., 2020) and are recommended for process intensification of mixing-sensitive operations (Myers et al., 2018).

1.1. Pressure drop in static mixers: Newtonian fluids

The pressure drop across a static mixer is often considered as the quantity of primary importance for the proper selection of a static mixer for a particular application and for the correct sizing of the pump for the inlet feed (Thakur et al., 2003). For an incompressible, isothermal fluid, the pressure drop can be related to the friction factor using

$$\Delta p = \frac{2f\rho u^2}{D}LN_e \quad (1)$$

For the flow of laminar fluids, the friction factor is related to the Reynolds number of an open pipe as $f = \frac{C}{Re}$, where $C = 16$ for laminar flow in an open pipe and $C > 16$ for static mixer inserts. Static mixer vendors typically report pressure drop correlations as a ratio with reference to the open pipe as $Z = f_{\text{mixer}}/f_{\text{pipe}}$ (Thakur et al., 2003). To report the friction factor directly, the constant C can be re-expressed as $C = 16Z$. Correlations of the friction factor f for various commercial static mixers over a wide range of Reynolds numbers are reported in (Thakur et al., 2003).

Some vendors alternatively report pressure drop data in terms of a power constant K_p . This choice of constant is more popular for stirred tank systems and can be obtained from $Ne \cdot Re = K_p$, where the Newton number, $Ne = 2f$. Comparison of vendor supplied values of K_p with estimates obtained using experiments and CFD simulation for different commercial mixers are reported in (Thakur et al., 2003).

1.2. Pressure drop in static mixers: non-Newtonian fluids

In the case of non-Newtonian inelastic fluids, the friction factor correlation for static mixers may be expressed as

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Nomenclature

u	superficial velocity m/s (= flow rate/cross-sectional area)
ρ	density kg/m ³
p	pressure Pa
D	pipe diameter m
L	length of mixing element m
N_e	number of mixing elements
u_m	mixer velocity m/s
D_h	hydraulic diameter of mixer m
ϵ	void fraction of mixer geometry
t	time
ω	frequency rad/s
λ	relaxation time s
G'	storage modulus Pa
G''	loss modulus Pa
G_0	elastic modulus Pa
η	kinematic viscosity kg/m/s
D	deformation rate tensor 1/s
W	rotation tensor 1/s
f, f_v, f_d	friction factor
$\dot{\gamma}_m$	mixer strain rate
Re	Reynolds number of pipe
Re_{mixer}	Reynolds number of mixer
Wi_{mixer}	mixer Weissenberg number = $\dot{\gamma}\lambda = N_1/\tau_{12}$
De	Deborah number = $\lambda u_m/L$
Ma_v	Viscoelastic Mach number = $u_m/\sqrt{\frac{\eta}{\rho\lambda}}$
El	Elasticity number = Wi_{mixer}/Re_{mixer}

$$f = 2 \left(\frac{K_L}{Re_m} + K_T \right). \quad (2)$$

In Eqs. (2,5,6) the friction factor correlation is expressed in terms of a mixer (or Generalised) Reynolds number defined as

$$Re_m = \frac{\rho u_m D_h}{\eta(\dot{\gamma}_m)}. \quad (3)$$

The measured constants K_L and K_T (valid for laminar and turbulent flows respectively) are specific for each mixer and dependent on scale and material of construction. For the SMX mixer these were determined to be 273 and 2 respectively. The viscosity value used in Eq. 3 is evaluated at a Metzner-Otto type average mixer shear rate $\dot{\gamma}_m = K_G u/D$ Hartt et al., (2017). K_G depends on the details of the mixer geometry and is supplied by the vendor; Streiff et al., (1999) reports values of K_G for various commercial static mixers. Here u_m represents the average velocity in the mixer taking into account the void fraction ϵ due to the presence of mixer elements i.e., $u_m = u/\epsilon$ Thakur et al. (2003). Correlations of friction factor with the Generalised Reynolds number for several shear thinning fluids modelled using the Cross model were reported in Li et al. (1997).

1.3. Viscoelastic flows in static mixers

1.3.1. Experimental studies

Investigations of viscoelastic flows through static mixers are scarce in the literature. Mixing of inelastic viscoplastic fluids using ERT techniques in SMX mixers by Jegatheeswaran et al. Jegatheeswaran et al., (2017) showed that the flowfield in the mixer promoted fluid stretching and rate of diffusion. Peryt-

Stawiarska & Jaworski (Peryt-stawiarska, 2012) report LDA measurements of velocity profiles in an SMXTM mixer. Their measurements reveal a loss of dynamic similarity of the profiles with Reynolds number in contrast to Newtonian fluids. Ramsay et al., (2016) investigated the mixing of viscoelastic fluids in a KenicsTM mixer and showed that mixedness decreased with increasing elasticity of the fluid. Shah & Kale (Shah and Kale, 1991) and Chandra and Kale, (1992) reported experimental measurements of the pressure drop through static mixers for Newtonian and non-Newtonian fluids including viscoelasticity. They showed that fluid elasticity leads to an increase in the pressure drop when compared to inelastic fluids. They proposed a modification to the inelastic friction factor correlation to account for viscoelastic effects through the introduction of the Weissenberg number, $Wi_{mixer} = N_1/\tau_{12}$ evaluated at the mixer shear rate $\dot{\gamma}_m$. This allows a friction factor for viscoelastic fluids to be obtained from the corresponding value for inelastic fluids (Eq. 2). This elastic correlation can be expressed as

$$f_v = f [1 + R(Wi_{mixer})^s] \quad (4)$$

Here R and s are empirically determined constants specific to a particular static mixer. For the Sulzer SMXTM Chandra and Kale (1992) report these values as $R = 0.21$ and $s = 1.12$ based on measurements on aqueous polymer solutions. Hartt et al. (2017) measured the pressure drop of viscoelastic worm like micellar (WLM) fluids in a Sulzer SMXTM mixer. They adopted an alternative approach whereby they correlated their friction factor data to Eq. 2 accounting for viscoelasticity of the fluids by using the Cox-Merz rule and a shear thinning viscosity model whose parameters were fit to the complex viscosity versus frequency.

1.3.2. Computational studies

The use of computational methods to investigate viscoelastic flows in continuous processes such static mixers are also scarce in literature as the bulk of studies have focussed on batch systems such as stirred tanks. However recent studies have demonstrated the potential for CFD methods in gaining insight into the detailed fluid dynamics and mixing performance of static mixers albeit for Newtonian (Zalc et al., 2003; Liu et al., 2006; Visser et al., 1999; Hirschberg et al., 2009) shear-thinning and viscoplastic fluids (Jegatheeswaran, 2016; Rahmani et al., 2006; Rahmani et al., 2007). CFD simulations were used in these studies to analyse the type of deformations in the flow field and distribution of deformation rates within the mixer and optimise mixer design based on the efficiency for mixing. Simulations by Rahmani et al. Rahmani et al., (2007) on shear-thinning and shear-thickening fluids showed that the SMX mixer is superior to helical mixers for mixing but at the cost of higher energy input requirements. Detailed investigations into the mixing performance of viscoplastic fluids in the SMX mixer was carried out by Jegatheeswaran et al. (2018). Chaotic advection caused by the mixer elements was found to be essential for the laminar mixing of such non-Newtonian fluids.

1.3.3. Overview

In this paper CFD methods are used for the first time to simulate the flow of viscoelastic fluids through a Sulzer SMXTM mixer. We use previously published results for model fluids as part of the validation of the pressure drop and present additional experimental pressure drop measurements of non-Newtonian (shear thinning and viscoelastic)

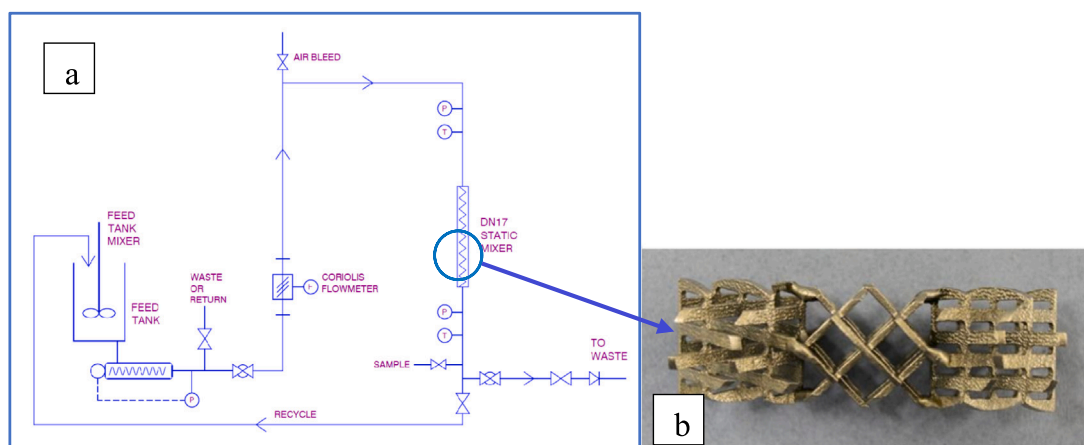


Fig. 1 – (a) Static Mixer Test Rig (b) Image of Sulzer SMX+™ DN17 static mixer (3 elements).

real process fluids flowing through the SMX+™. The real process fluids are taken from different classes of fluid than the model fluids typically used in such studies. The first fluid is a high phase volume emulsion as typified by a mayonnaise (Dubbelboer et al., 2016). The second is a surfactant structured system of a three-dimensional lyotropic liquid-crystal matrix of elongated and entangled flexible self-assembly structures. Examples include shampoo, drilling muds and drug delivery systems and display a range of viscoelastic properties (Dreiss and Feng, 2017). The experimental and CFD results are compared against the empirical correlation of Chandra and Kale (1992) as given by Eq. 4. This empirical model only relies on the rheological characteristics of the fluids not the underlying microstructure and these characteristics are presented in the next section for the process fluids mayonnaise and shampoo. Finally, we examine through the CFD the details of the flow behaviour in static mixers to understand better the pressure drop measurements and potential impact on mixing behaviour.

2. Experimental methods

2.1. Rheology methods

A Malvern Gemini 2 Stress Controlled Rheometer was used for viscometric and oscillatory measurements. The principal measurement system used was the 40 mm diameter 4° cone and plate (CP4/40). Basic flow characterisation was conducted using both stepped shear rate and ramped shear rate tests for each fluid sample. The shear rate ranges were chosen to span the pipe and mixer shear rates experienced by the fluids over the flowrate range tested (1 to 600 s⁻¹). Typically, lower shear rates (to 0.01 s⁻¹) were also included for the stepped rate tests. Sample yield stress was assessed using stress ramp tests. This involved applying a gradually increasing stress and monitoring the instantaneous viscosity for an inflexion of the curve i.e., the onset of flow.

Test fluid viscoelastic properties were measured using oscillatory techniques. The technique applied a stress or

strain whose value was changing continuously according to a sine wave equation. The induced response (strain or stress) also followed a sine wave. The sample was continuously excited but never exceeded a certain strain and hence sample structure was not destroyed (providing steps were taken to keep the strain small enough). Amplitude Sweeps were carried out on each sample to verify the upper strain limit of the Linear Viscoelastic Region (LVER) by oscillating at a fixed frequency and slowly increasing the applied amplitude (strain or stress). A strain value within the LVER was selected for each frequency sweep. The combination of viscous and elastic components in a material respond in different ways depending on the 'speed' at which you try to move it. By oscillating over a wide range of frequencies we obtained the characteristics over a range of time scales as high frequencies relate to short times whereas low frequencies relate to long time scales.

Rheology measurement temperature was set as closely as possible to the mean of the static mixer inlet and outlet values measured during the pressure-drop experiments.

Two test fluids were supplied by Unilever:

1. Mayonnaise,
2. Shampoo.

Two 100 ml samples of each test fluid were taken at the time of the pressure-drop measurements. The first 'A' sample was from the feed tank before the fluid had passed through the pump or static mixer, the second 'B' sample was taken from the sample tap downstream of the static mixer. Both 'A' and 'B' rheology results are presented in Section 4.1 for each test fluid.

2.2. Pressure drop method

The static mixer test rig (Fig. 1a) consisted of the following features: a 0.05 m³ capacity feed tank mounted directly above a variable speed progressive cavity pump, (Netzsch Nemo NM015BY, 0.75 KW motor, 14-360 l/h at up to 12 bar

Table 1 – Key dimensions of the SMX+ mixer at DN17 scale.

DN (mm)	Pipe diameter D [mm]	Hydraulic diameter D _h [mm]	Diameter ratio D _h /D [-]	Number of crossing bars over height n [-]	Number of parallel bars over length N _p [-]	Number of crossing bars over width N _x [-]	Voidage ε [-]
17	18	6.8	0.378	2	3	6	0.9

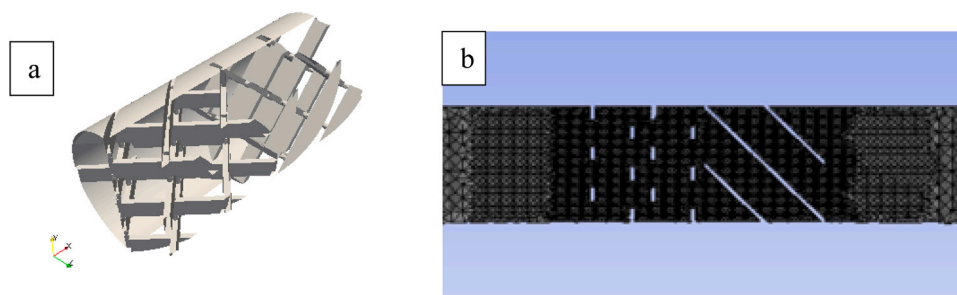


Fig. 2 – (a) Cut-away of the mixer element region of the CAD model of the static mixer, (b) Cross-section showing grid refinement regions near the mixer element surfaces.

differential pressure) used to deliver the bulk liquid to the mixer, a calibrated Endress & Hauser Promass 83 Coriolis flowmeter with an accuracy of $\pm 0.5\%$ of measured value over the range 0–4000 kg/h, upstream and downstream WIKA S-10 pressure transducers (0–10 bar, accuracy $\leq \pm 0.5\%$ span) and up and downstream thermocouples.

The DN17 scale Sulzer SMX+TM static mixer with 24 elements was tested in this study. This was found to be a reasonable scale for making pressure and flow measurements, avoiding blockages, and minimising the consumption of raw materials and production of waste. This choice also afforded manageable size of pumps, tanks and pipework whilst operating in laminar flow. The mixer elements were housed in a 18 mm pipe and the test section was orientated for vertical down flow. An image of the mixer elements is shown in Fig. 1b. The hydraulic diameter is $D_h = 6.88$ mm, with a pitch ratio of 1 so that the ratio $L_{\text{mixer}}/D = 24$. Other key dimensions of the mixer elements are given in Table 1.

The method comprised of a series of discrete tests at increasing pump speeds until either maximum pump speed (50 Hz) or maximum rig pressure (10 bar) was reached. One set of tests was completed with flow going once through to waste ('A' tests) and one set of tests with flow recirculating back to the feed tank ('B' tests). Each test comprised of a single pump speed, each test commenced at 0 Hz, stepped up to the chosen speed, held the chosen speed for a couple of minutes and returned to 0 Hz. For each test, raw pressure data from each pressure transducer, T and mass flow data were logged at a rate of 100 Hz and stored in the DAQ software. Any pulsatile fluctuations were recorded, and the presented data was based on time average over 10 s+. It was found that pulsations were a small ($< 1\%$) component of even the lowest flow measurements.

3. CFD modelling

3.1. Governing equations and rheological model

The flow of an incompressible, isothermal fluid is described by the momentum and continuity equations as

$$\frac{\partial u}{\partial t} + (u \cdot \nabla)u = -\nabla p + \nabla \cdot \sigma, \quad (5)$$

$$\nabla \cdot u = 0. \quad (6)$$

The stress tensor σ is decomposed into a solvent and polymer (extra) stress contributions using

$$\sigma = 2\eta_s D + \sum_{i=1}^n \tau_{pi}. \quad (7)$$

Here, the assumption is that elastic stresses built by deformation of the fluid relaxes over a spectrum of n relaxation times, λ_i , and τ_{pi} represents the contribution of the i^{th} relaxation mode to the total extra stress tensor. In the small strain limit, the viscoelastic behaviour of the fluid is linear and can be represented in terms of its response to a small amplitude oscillation of frequency ω . The response can be modelled as a spectrum expressed as a superposition of the n relaxation modes as

$$G' = \sum_{i=1}^n \frac{G_{0i}(\lambda_i \omega)^2}{1 + (\lambda_i \omega)^2}, \quad (8)$$

$$G'' = \sum_{i=1}^n \frac{G_{0i}(\lambda_i \omega)}{1 + (\lambda_i \omega)^2} \quad (9)$$

The linear viscoelastic spectrum from the experimental oscillatory measurements is fit to Eqs. 8 and 9 using a non-linear least squares method. For the CFD simulations surfactant solutions have also been considered. The rheological behaviour of these fluids have been investigated in previous studies (Helgeson et al., 2009; Ober et al., 2011). For such solutions the linear viscoelastic spectrum is described by a single relaxation mode.

For strains beyond the linear limit, differential constitutive models can be used to describe the non-linear rheology. In this case, the extra stress tensor is described in a coordinate system that stretches and rotates with the fluid. For the simulations in this paper, the Giesekus constitutive model (Giesekus, 1983; Giesekus, 1982) is used to describe the transport and stress relaxation of the i^{th} relaxation mode. This model can be expressed as

$$\frac{\partial \hat{\tau}_{pi}}{\partial t} = \frac{1}{\lambda_i} \left(-\tau_{pi} - \frac{\alpha_i \lambda_i}{\eta_{pi}} \tau_{pi}^2 \right) + 2 \frac{\eta_{pi}}{\lambda_i} D. \quad (10)$$

This model has been used in several previous studies to describe steady state rheology of solutions of polymers melts, worm-like micelles and aqueous gum solutions (Helgeson et al., 2009; Köpplmayr and Miethlinger, 2014; Germann et al., 2016; Pourjafar and Sadeghy, 2012; Torres et al., 2014). It has the advantage that analytical solutions are available for the viscometric functions in simple extension and shear (Giesekus, 1982) that can be used to fit to experimental data. The derivate (represented as top hat) in Eq. 10 represents the upper convected (or Oldroyd) time derivative. In the framework of the Giesekus model, the anisotropy mobility parameter α represents the strength of the coupling between the polymer chains (or surfactant micelle) with the flow (Bird et al., 1987). The Giesekus model can predict shear thinning viscosity, first and second normal stress differences and bounded elongational viscosity (Giesekus, 1982; Bird

Table 2 – Summary of measured rheological properties.

Test Fluid	Temperature (C)	Flow Curve Parameters				Stress Ramp			Normal Stress Fit			Oscillatory Parameters			
		τ_y (Pa)	K	n	R^2	τ_y (Pa)	m	c	$N_1=mY+c$	R^2	Wi_{mixer} [-]	G' Pa	G'' Pa	Phase Angle (°)	LVER Strain Units
		Shear rate (s ⁻¹)	Density (Kg/m ³)												
Mayonnaise A	13.5	39.5	17.88	0.44	0.9982	19	0.48	0.99	0.994	0.994	0.3-0.9	530	63	6.8	0.02
Mayonnaise B	13.5	38.3	15.32	0.45	0.9961	17.5	0.486	170	0.939	0.939	1.3-1.5	480	56	6.7	0.01
Shampoo A	10.5	51.8	0.35	0.74	0.948	18.5	10.39	-2.44	0.999	0.999	10-69	180	88	26.1	0.1
Shampoo B	10.5	46.1	2.26	0.42	0.9	17.5	8.57	25.7	0.999	0.999	9-67	160.5	92.2	29.9	0.1

Table 3 – Experimental conditions for the pressure measurements for Mayonnaise.

Flowrate	Velocity	Shear Rate	Re _{mixer}
kg/h	(m/s)	(1/s)	-
44.26	0.06	119.19	0.25
65.30	0.09	175.85	0.47
87.24	0.12	234.93	0.76
104.89	0.15	282.45	1.03
130.90	0.18	352.50	1.47
173.63	0.24	467.58	2.33
216.56	0.30	583.19	3.33

Table 4 – Experimental conditions for pressure measurements of Shampoo.

Flowrate	Velocity	Shear Rate	Re _{mixer}
kg/h	(m/s)	(1/s)	-
22.92	0.03	56.88	0.19
47.52	0.06	117.92	0.76
72.85	0.09	180.76	1.66
98.10	0.13	243.43	2.84
123.26	0.16	305.86	4.27
147.93	0.19	367.08	5.88
172.72	0.22	428.58	7.69
197.94	0.25	491.17	9.71
222.41	0.29	551.87	11.83
247.52	0.32	614.19	14.16

et al., 1987). For the purposes of the computational simulations, the steady shear data from measurements is fit to the analytical solutions using non-linear least squares to obtain values for Giesekus parameter α_i . During the fitting process, the values of the relaxation time (λ_i) and viscosity ($\eta_{pi} = G_{0i} \lambda_i$) are fixed to the previously fitted values obtained from the linear viscoelastic data.

3.2. Geometry and mesh development

The DN17 and DN50 SMX+™ static mixer geometries were considered for the computational simulations. Owing to the high computational cost of the viscoelastic flow simulations, the geometry was restricted to two mixer elements. A CAD geometry was created to represent the computational domain comprising an outer pipe with two mixer elements oriented 90° to each other (Fig. 2a). An unstructured tetrahedral mesh was developed to discretise this complex computational domain. The grid resolution was refined in the regions near to the static mixer surfaces (Fig. 2b). Typical grid sizes ranged from 1.1 million to 3.46 million cells for the different mixer geometries. Grid independence tests based on relative change in pressure drop and validation against Newtonian flow friction factor benchmark data were carried out (see Appendix 1 for details).

3.3. Numerical methods

CFD solutions were carried using the rheoTool library (Pimenta and Alves, 2017) of the OpenFOAM open source CFD toolbox (Weller et al., 1998). This library has been extensively validated against several benchmark flows of viscoelastic fluids (Pimenta and Alves, 2017) and such solvers have been used in several studies on the flows of polymer melts and surfactant solutions, mostly in microfluidic geometries

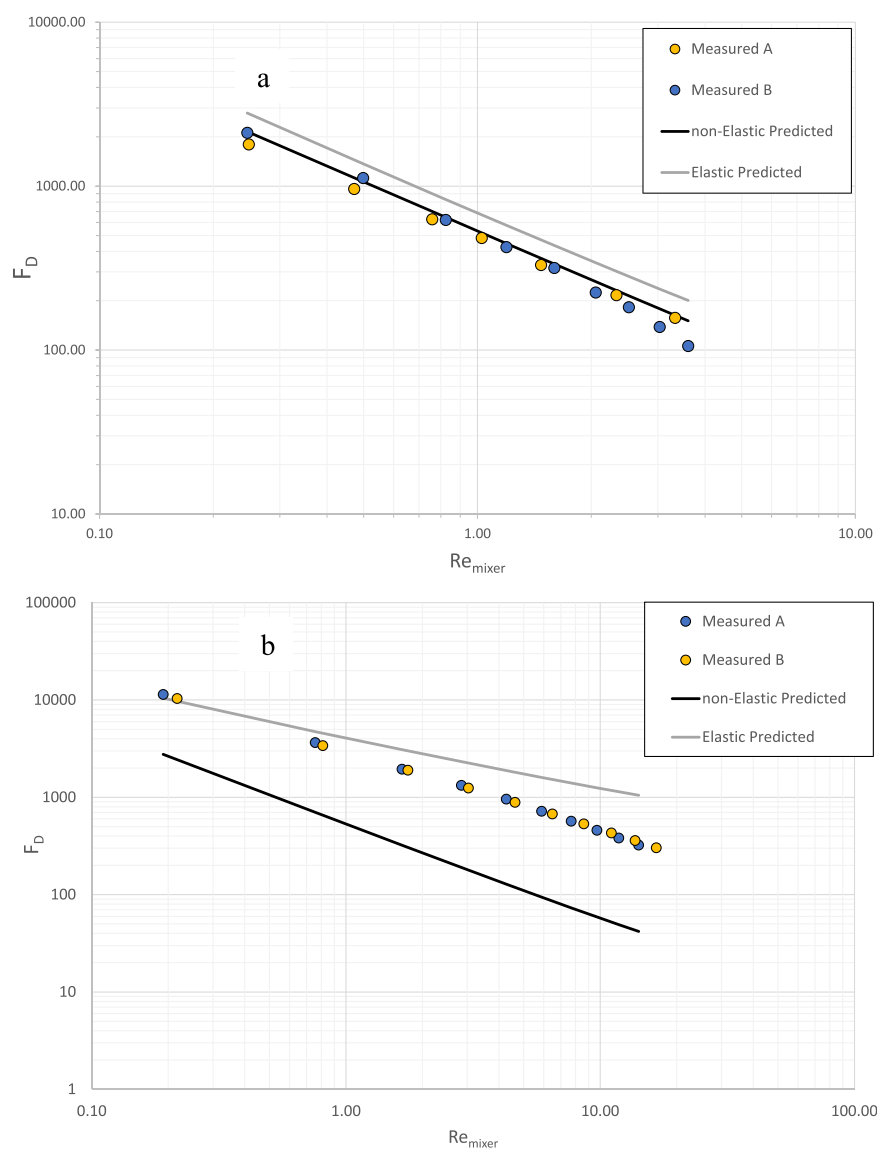


Fig. 3 – (a) Measured friction factor vs literature correlations (Eqs. 2 and 4) for Mayonnaise ‘A’ and ‘B’ samples and (b) Shampoo ‘A’ and ‘B’ samples. The elastic correlation (Eq. 4) was evaluated using constants reported in Chandra & Kale (Chandra and Kale, 1992).

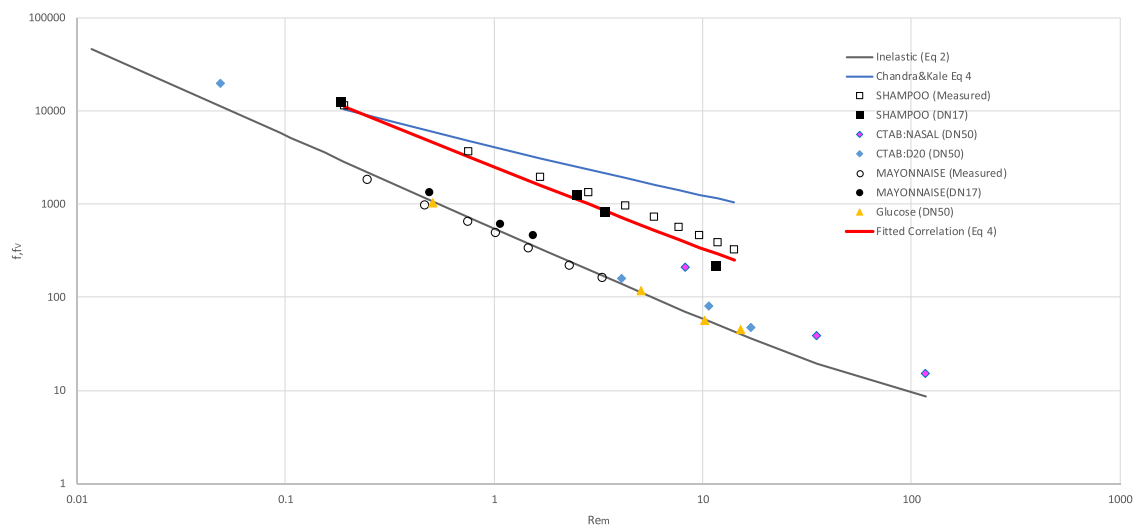


Fig. 4 – Friction factors for various fluids. CFD results are denoted by solid symbols and measurements by open symbols. The inelastic correlation (Eq. 2) is displayed using a solid black line and Eq. 4, with $R=0.21$, $s=1.12$ from Chandra & Kale (Chandra and Kale, 1992), using solid blue line. A fit of Eq. 4 to the Shampoo data ($R = 0.267$, $s = 1.57$) is displayed using solid red line. (For interpretation of the references to colour in this figure, the reader is referred to the web version of this article.)

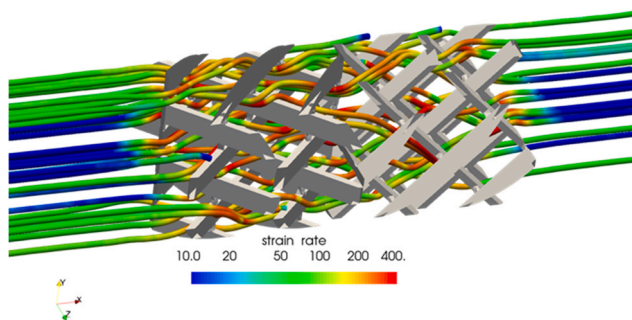


Fig. 5 – Flow streamlines coloured by the strain rate (1/s) for the flow of shear thinning fluid.

(Hooshyar and Germann, 2019; Davoodi et al., 2019; Abuga and Chinyoka, 2020; Kalb et al., 2018; Buel et al., 2018).

OpenFOAM employs the finite volume method for discretisation of the governing equations. The pressure and velocity field are calculated using the SIMPLEC algorithm (Van Doormaal and Raithby, 1984). Convective terms are discretised using high resolution schemes (HRS) handled with a deferred correction approach. Transient terms are discretised using first order implicit scheme. The time-step was selected to satisfy the Courant-Friedrichs-Lewy condition (Ferziger and Perić, 2002), defined as $\Delta t < \Delta x^2 / (2_\rho^2 + U\Delta x)$. This expression relates the time step to the characteristic time scale for both diffusion and convection and is evaluated for a typical cell in the refined region near mixer element surfaces. Steady-state convergence of the solution was determined when the pressure difference between the inlet and outlet faces reached an asymptotic value.

Numerical solutions of viscoelastic flows are challenging due to the so-called High Weissenberg number problem (Keunings, 1986) wherein the numerical schemes breakdown when a critical Weissenberg or Deborah number is exceeded. This problem has been recognised as arising due to the steep exponential growth of the extra stress tensor in response to deformation and convection, particularly in regions of large strain rates or stagnation points. The Log Conformation Reformulation approach (Fattal and Kupferman, 2004) has been proposed to overcome this problem and is adopted for the simulations carried out in this study. The numerical schemes are also additionally stabilised using the Both Sides Diffusion method (Pimenta and Alves, 2017).

The mean velocity (selected to match the flowrates from the experiments) was specified at the pipe inlet, and an outlet boundary condition (atmospheric pressure) was specified at the pipe outlet. No-slip velocity conditions were specified on the pipe walls and mixer element surfaces. The extra-stress tensor was specified as zero at the pipe inlet, zero-gradient at the pipe outlet and linearly extrapolated from the boundary cells to the boundary faces on all walls (Pimenta and Alves, 2017).

4. Results and discussion

4.1. Rheology results

Table 2 provides a summary of the rheological test results. All the test fluids exhibited shear-thinning properties with Herschel-Bulkley parameters fitting the flow-curve data best over the static mixer shear-rate range. The Herschel-Bulkley yield stress values for a given fluid were consistently higher

than those determined from stress ramps. Shampoo and Mayonnaise both gave H-B yield stresses of ≈ 40 –50 Pa and stress ramp values of 17.5–19 Pa.

All test fluids showed varying degrees of viscoelastic behaviour. Shampoo exhibited by far the largest normal forces under the conditions in the static mixer, resulting in $Wi_{\text{mixer}} = 10$ –70, which was significantly greater than Mayonnaise $Wi_{\text{mixer}} = 0.3$ –1.5 values. Further details of the fluid rheology are given in Appendix 2.

4.2. Pressure drop measurement results

Two sets of pressure drop results were measured for each test fluid, ‘A’ with the test fluid pumped through the mixer straight to waste and ‘B’ with the test fluid returned to the feed tank (see Fig. 1). Re_{mixer} and Wi_{mixer} were calculated using ‘A’ rheology sample flow curve parameters for straight to waste ‘A’ tests, whilst ‘B’ rheology sample flow curve parameters were used for recirculated ‘B’ tests. The experimental conditions pertaining to these measurements are listed in Table 3 and Table 4 for the Mayonnaise and Shampoo respectively. Pressure drop for once through to waste (A) and recirculated tests (B) F_D were very similar when combined with pre- and post-mixer rheology samples respectively. Non-elastic F_D prediction using Eq. 2 accurately predicted Mayonnaise F_D (Fig. 3a) However, non-elastic F_D prediction massively under-predicted measured Shampoo F_D by almost an order of magnitude (Fig. 3b). Elastic F_D prediction using Eq. 4 with Chandra & Kale (1992) constant and exponents more accurately predicted Shampoo F_D .

The maximum instrument errors were Gemini Rheometer: viscosity $\pm 5\%$ measured value, flowmeter: $\pm 0.5\%$ measured value and pressure transducers: ± 50 mbar (which equated to $\pm 5\%$ to $\pm 1\%$ measured value over the range of upstream and downstream pressures). The maximum errors in friction factor were therefore $\pm 10.5\%$ to $\pm 4.5\%$ over the operating range with the largest maximum error associated with the lowest flowrate conditions. The maximum error in Reynolds number was $\pm 5.5\%$.

4.3. CFD results

The CFD simulations were conducted at a range of representative flowrates for both DN50 and DN17 mixer geometries. The mixer Reynolds number and Weissenberg number at each flowrate was determined using the expressions described in Section 1.3. The Reynolds number ranged between $O(10^{-1})$ to $O(10^2)$ so that the flow regime in the mixer was laminar (in SMX mixers secondary flows are expected when $Re > 100$ (Migliozi et al., 2021)). The Elasticity number El defined as the ratio of elastic to inertial forces (Wi/Re) for the viscoelastic simulations was typically $O(\text{Peryt-stawiarska, 2012})$ corresponding to flow regimes where elasticity dominates inertial effects.

For viscoelastic flows, a measure of unsteadiness in the flow can be ascertained using the Deborah number De , defined as the ratio of the longest relaxation time of the fluid to the characteristic time scale of the deformation in the flow (Poole, 2012). Taking the residence time in a single mixing element, L/u_m , as the characteristic flow time, the Deborah number $De = \lambda / (L/u_m)$ was less than 1; the flow through the mixer is thus expected to remain steady.

Recently elastically induced flow instabilities were reported in static mixers (Migliozi et al., 2021). The onset of such instabilities is characterised by a viscoelastic Mach

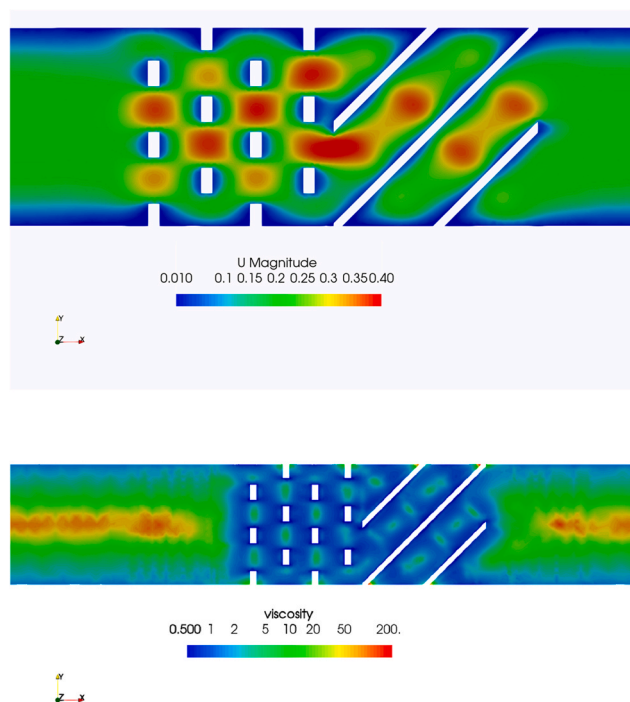


Fig. 6 – Longitudinal section of the mixer showing distribution of velocity (m/s) (top) and viscosity (Pa.s) (bottom) for the shear thinning fluid.

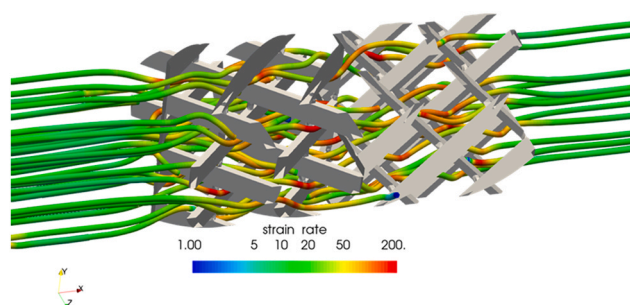


Fig. 7 – Streamlines for the flow of viscoelastic fluid through the mixer, coloured by the magnitude of strain rate (1/s).

number, Ma_v , defined as the ratio of the characteristic flow velocity in the mixer, u_m , to characteristic velocity of the elastic waves, $\sqrt{\eta/\rho\lambda}$ Ahrens et al., (1987). In this study, Ma_v was typically < 1 , thus such instabilities are expected to be absent for the viscoelastic flows considered here.

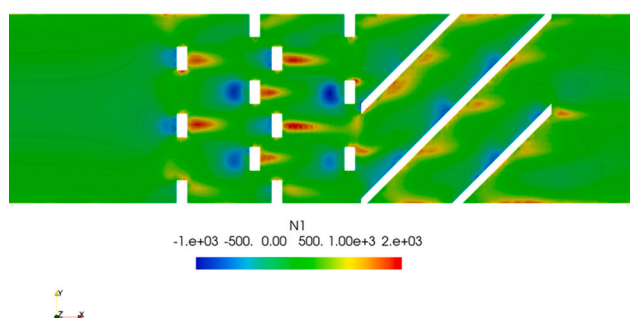


Fig. 8 – Longitudinal section of the mixer showing distribution of the first normal stress difference (N).

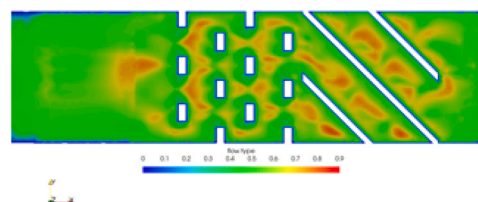


Fig. 9 – Longitudinal-section of the mixer region showing classification of the flow field using Eq. 11.

The simulations were run using the High-Performance Computing cluster at the University of Manchester. Typical job sizes for the viscoelastic simulations ranged from 72 to 96 processor cores and required on average 60 h for completion. Once the numerical simulations attained convergence to steady state, the difference in area averaged pressure between the beginning and end cross-sections of the mixer elements was used to evaluate the friction factor (using Eq. 1).

4.3.1. Pressure drop predictions

Pressure drop predictions for the process fluids (Mayonnaise and Shampoo) using CFD simulations are reported here. In addition, two model surfactant solutions, (16.7% CTAB- D_2O and 30-240 mM CTAB-NaSal) from literature (Helgeson et al., 2009; Ober et al., 2011) were also considered for the simulation. For the mixer conditions considered here, Mayonnaise could be modelled as a viscoplastic material as the experimental measurements indicated that $Wi_{\text{mixer}} < 1$. The Herschel-Bulkley equation was used to describe the non-Newtonian behaviour. Any changes in droplet size of the Mayonnaise were neglected as maximum shear rates in the mixer (1000 1/s) corresponding to typical flow rates were not high enough to cause droplet breakup. The mean droplet size of a typical mayonnaise emulsion was found to be below the maximum stable droplet size for dense emulsions at these shear rates (obtained using the model of Dubbelboer et al. Dubbelboer et al., (2016)).

The CFD results are plotted in Fig. 4 using solid symbols whilst the experimentally measured values for the process fluids are displayed using open symbols. The Chandra & Kale (Chandra and Kale, 1992) correlation and the Newtonian (inelastic) correlations are displayed using solid lines. The CFD predictions for Mayonnaise (modelled as a shear-thinning fluid) are close to the experimental data and inelastic correlation. The friction factors for all the viscoelastic fluids are higher compared to the inelastic correlation. There is good agreement between the measured and CFD values for Shampoo particularly at lower Re numbers. The CFD results give a better prediction of the friction factor values compared to the Chandra & Kale correlation. The Chandra & Kale correlation tends to over-predict the friction factor values; the predictions are closer for the more strongly viscoelastic Shampoo (in which the Wi number is relatively high, O(Perytstawiarska, 2012)) and at low Re number. The CFD and measured Shampoo data is fit to Eq. 4 to obtain revised constants (solid red line in Fig. 4) $R = 0.267$ and $s = 1.57$.

4.4. Flow field analysis

Pressure drop estimations provide a gross measure of the static mixer performance. It is essential for proper selection of a mixer for a particular application and sizing of pumps

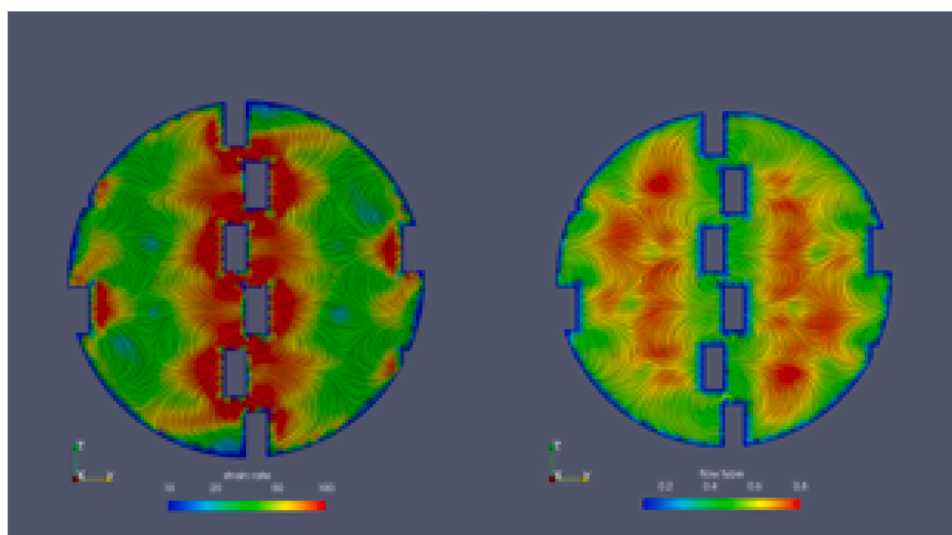


Fig. 10 – Comparative view of a cross-section through the crossbars showing the flow field classification (right) and distribution of strain rate (1/s) (left).

and feeds (Thakur et al., 2003). However successful application of static mixers for process intensification requires a more detailed understanding of the flow dynamics. For instance, better mixing performance may be achieved through

an intensification of fluid deformation in the mixer (Jegatheeswaran et al., 2018). CFD methods are a valuable investigative tool that may be used to explore the details of the flow field and deformation history within the mixer. For

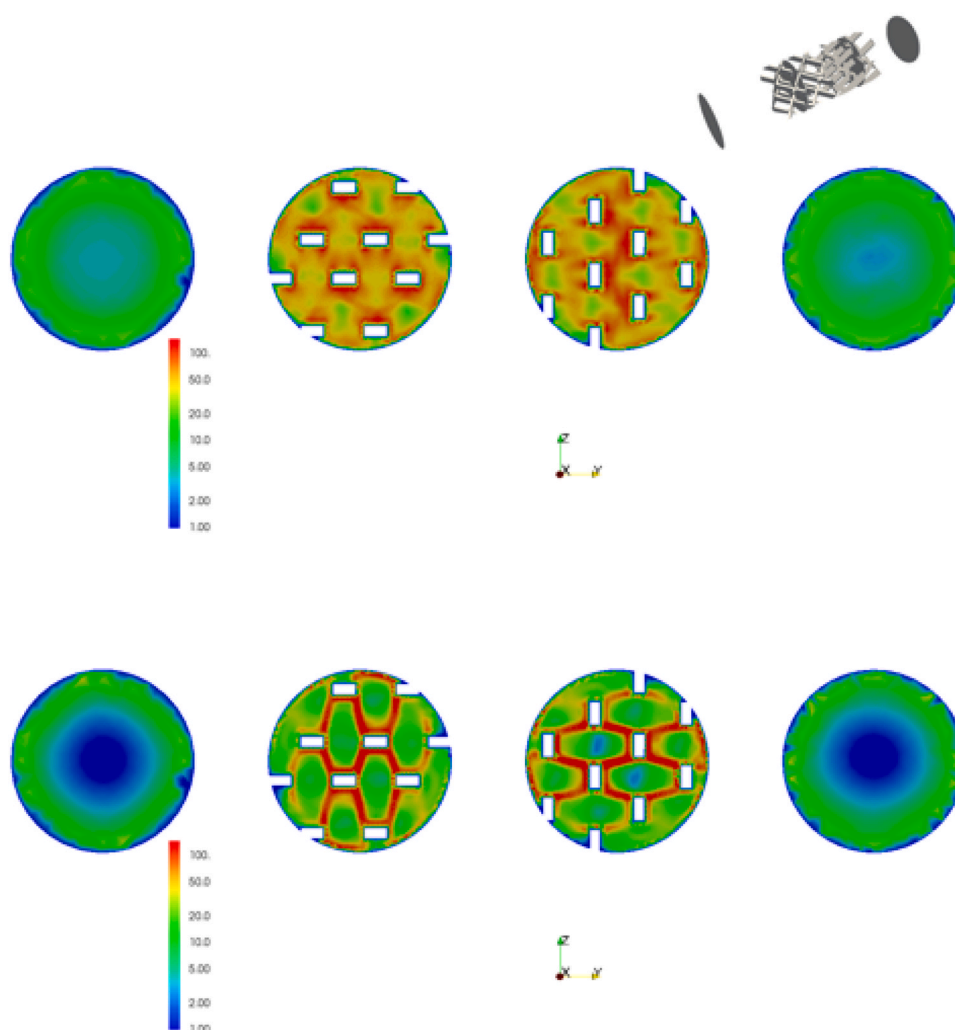


Fig. 11 – Distribution of strain rate (1/s) over different cross-sectional planes in the domain for the viscoelastic fluid (top) and inelastic fluid (bottom). The locations of the planes are shown in the inset.

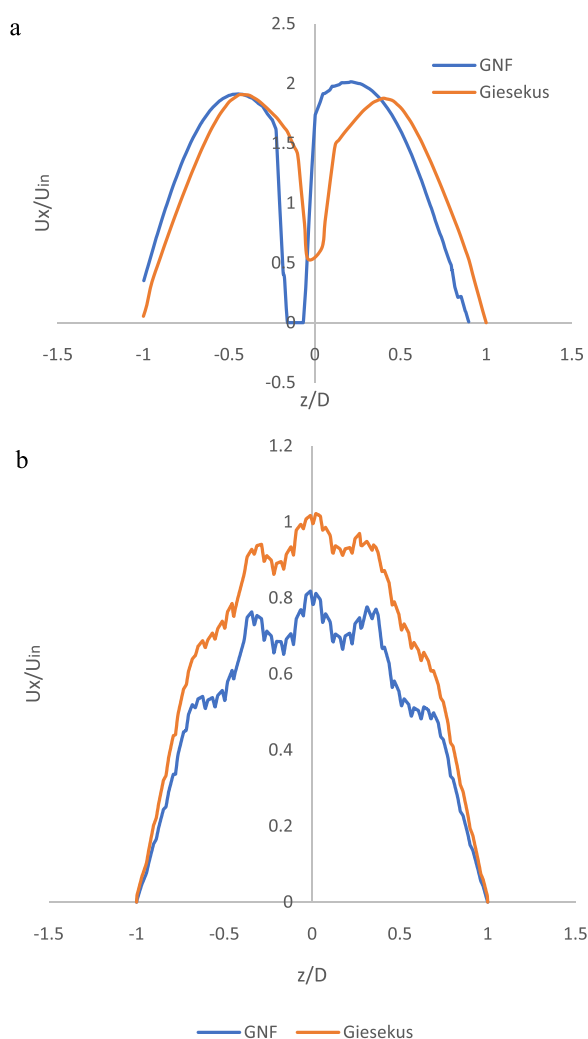


Fig. 12 – Plot of axial velocity (normalised by inlet velocity) across transverse axis for the elastic and inelastic fluids at axial location (a) through the mixer elements and (b) immediately downstream of the mixer elements.

viscoelastic fluids in particular, a precise understanding of the contribution of fluid elasticity to the mixer performance is lacking. Viscoelastic fluids are characterised by the presence of normal stress differences, but the contribution of these normal stress differences to mixing is still under debate in the literature (Ramsay et al., 2016). Some studies on stirred tank reactors show that secondary flows induced by the normal stresses aids convective mixing (Stobiac et al., 2014; Fradette et al., 2007; Ohmura et al., 2018), whilst other studies show that the effects are detrimental due to cavern formations and solid-like portions in the fluid domain being less readily mixed (Ramsay et al., 2016; Ramsay et al., 2016; Kolano and Kraume, 2019).

4.4.1. Shear thinning fluid

The flow-field for the flow of Mayonnaise (modelled as a shear thinning fluid) is analysed. Typical flow streamlines through a longitudinal cross-section of the mixer are shown in Fig. 5. The streamlines are coloured by the magnitude of the rate of deformation tensor D , denoted as the strain rate $\dot{\gamma} = \sqrt{2D:D}$. This quantity characterises the magnitude of the local straining of the fluid generated as it passes through the mixer element regions. The cutting and folding action of the fluid material lines by mixer crossbar elements generates

regions of maximum strain rate. The flow accelerates around the edges of the mixing elements, where the highest strain rates are generated. The ‘X’ shaped arrangement of the crossbar elements deflects the flow streamlines perpendicular to the axial flow direction, a characteristic of SMX mixers that helps to increase the velocity in the regions near the pipe wall (Hirschberg et al., 2009). Upstream and downstream of the mixing elements a core plug flow region of low strain rates can be observed in which the fluid remains unyielded. This is corroborated in Fig. 6 where the regions of high viscosity in the centre of the pipe can be observed.

4.4.2. Viscoelastic fluids

Shampoo was modelled as a viscoelastic fluid; the resulting Wi_{mixer} was $\sim O(1)$ (Peryt-stawiarska, 2012). The linear rheology could be described using a single-mode Maxwell equation. The nonlinear rheology was described using the Giesekus equation. Typical flow streamlines passing through the mixing element are shown in Fig. 7. The laminar nature of the flow can be observed and the fluid streamlines through the mixer region represent a streamwise tortuous flow akin to flow through porous media (Hartt et al., 2017). There is an absence of flow recirculation loops perpendicular to the main flow direction which have been observed for example in the viscoelastic flow through the Kenics mixer (Ramsay et al., 2016) and in-house CFD simulations). The streamlines are coloured by the strain rate magnitude.

To gain insight into the viscoelastic nature of the flow field the distribution of normal stress differences in the domain is analysed. These stress differences are associated with tension or compression in the flow streamlines and cause the microstructure to stretch or contract in the direction of the straining (Hinch, 2011). The distribution of the elastic stresses (depicted using the first normal stress difference, N_1) is shown in Fig. 8. The deformations in the flow induced by interaction with the mixer element surfaces gives rise to the regions where the elastic effects are most pronounced.

4.5. Flow field classification and mixing performance

Dispersive mixing is one of the key mechanisms of laminar mixing in static mixers particularly for highly viscous fluids (Tadmor and Gogos, 1979). It is generally accepted that elongational flow is more effective than simple shear flow for dispersive mixing (Bigio and Conner, 1995; Erwin, 1978; Connelly and Kokini, 2004). This is because the presence of non-zero vorticity in the shear flow can cause fluid clusters to rotate as a rigid body while being sheared. There is no rotation in pure elongational flow and thus the stretching deformation is effectively transferred to the fluid clusters causing them to pull apart and rupture into smaller segments. The velocity gradient tensor is a useful tool to classify the elongational or shear nature of the flow field and thus provide a measure of the efficiency for dispersive mixing by the flow field. The velocity gradient tensor can be decomposed into its symmetric D and anti-symmetric W parts and used to define a flow type classification index (Cheng and Manas-Zloczower, 1990; Manas-Zloczower, 1997),

$$\lambda_{\text{FlowType}} = \frac{|D|}{|D| + |W|}. \quad (11)$$

This index characterises the extent of elongational and rotational components in the flow field with values ranging

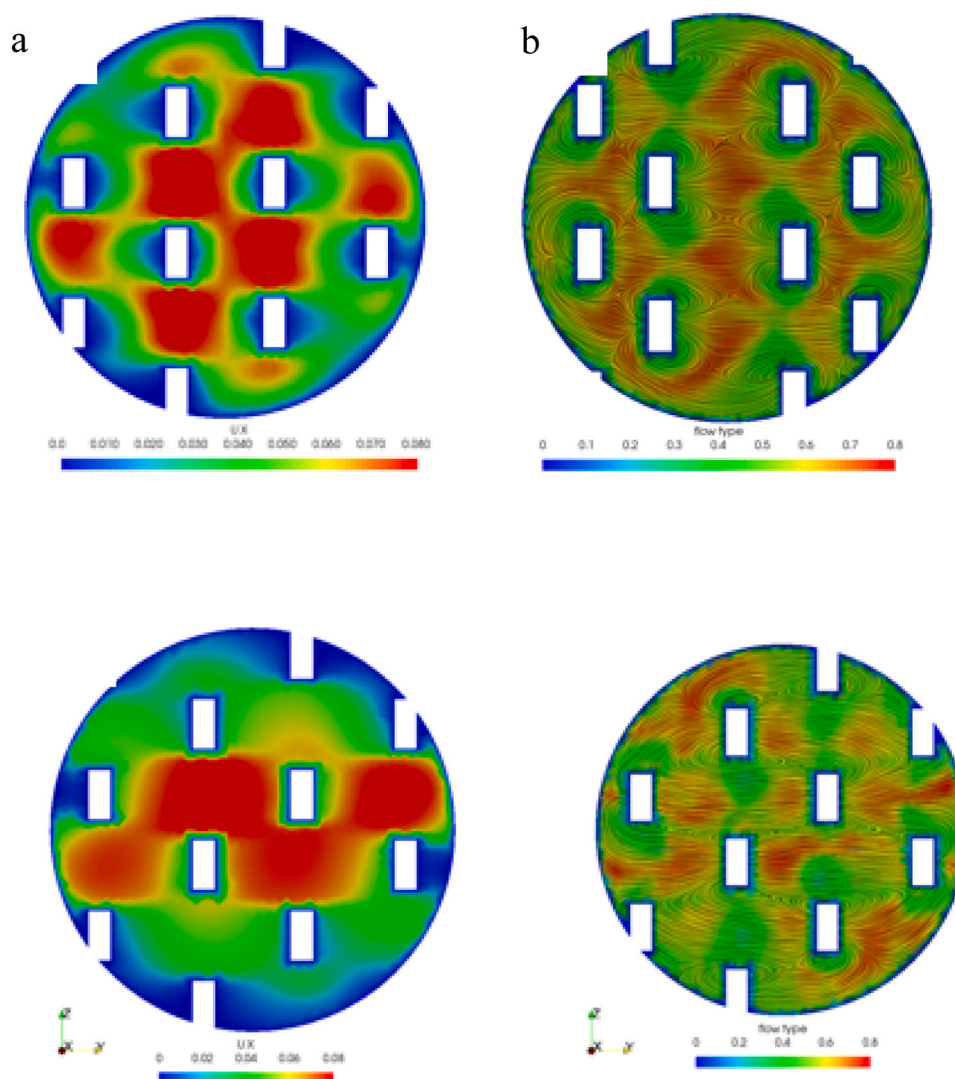


Fig. 13 – Cross-sectional plane through mixer element showing distribution of (a) axial velocity (m/s) and (b) flow-field classification and flow streamlines for the viscoelastic fluid and (c) axial velocity (m/s) and (d) flow-field classification for the inelastic fluid.

from 0 for pure rotation, 0.5 for simple shear and 1 for pure elongation. In Fig. 9 the bulk of the flow field through the mixer is that of pure shear flow ($\lambda_{FlowType} \sim 0.5$). The presence of the crossbar elements introduces regions dominated by elongational flow ($\lambda_{FlowType} > 0.8$) which are more favourable for dispersive mixing. The streamlines in these elongational regions are strain dominated and manifest as hyperbolas. It can be noted that the regions of strongest elongational flow (such as near the centre points of the hyperbola) do not always correspond to the peak values of strain rate (Fig. 10). Thus, the most favourable regions for dispersive mixing may not coincide with regions of strongest shear stresses and shear thinning. Processes that are shear dependent such as blending of high viscosity fluids into low viscosity, deagglomeration, droplet breakage etc. will favour regions closer to the mixer element surfaces where the flow field is pure shear dominated.

4.5.1. Comparison between inelastic and elastic fluids

In this section, the results of a simulation carried out for a shear-thinning inelastic fluid is compared with that for the viscoelastic fluid to ascertain the differences between the two types of fluids. The shear thinning fluid is modelled as a

fictitious GNF whose rheology is modelled using a Carreau-Yasuda equation fit to match the steady-shear rheology of the viscoelastic fluid. The simulations are run at the same inflow rate to match the mixer shear rate $\dot{\gamma}_m$ and Reynolds numbers.

The distribution of the strain rate over four cross-sectional planes in the domain are represented in Fig. 11 for the viscoelastic fluid. The interaction of the flow with the mixer crossbar elements generates higher strain rates compared to the upstream and downstream locations of the empty pipe; zones with highest strain rates are located around the crossbars and near the pipe walls. Assessing the distribution of strains rates in the domain for the shear thinning fluid (Fig. 11 bottom), the strain rate distribution is more non-uniform with localized zones of highest strain rates. Fluid elasticity thus serves to homogenise the strain rates over the cross-section through both the mixer zones and the empty pipe.

The instantaneous axial velocity distribution at a location through the mixer element and downstream of the mixer elements are plotted in Fig. 12 for the elastic and inelastic fluids. The velocity profile is more symmetric around the centre-axis for the elastic fluids. Downstream of the

elements in the empty pipe section, the profile is symmetric for both fluids; the peak inlet velocity is recovered for the elastic fluid compared to the GNF. In Fig. 13 the flow field on the entire cross-sectional plane through the mixer elements is represented. By comparing the distribution of the axial velocity of the GNF (Fig. 13a) with that of the viscoelastic fluid (Fig. 13c) it can be seen that the spread of the velocity in the cross-stream direction is lower in the GNF. The enhancement of cross-stream convective motions in the viscoelastic fluid may be attributed to the presence of the secondary normal stress differences (Debbaut et al., 1997; Yue et al., 2008). Fig. 13b and Fig. 13d show the flow field classification and flow streamlines for the viscoelastic fluid and GNF respectively. The flow streamlines between the crossbar elements is more parallel and the strain dominant regions of the flow field are smaller when compared to the viscoelastic fluid. Thus, regions where the flow is strongly elongational in nature is reduced in the GNF fluid when compared to the viscoelastic fluid.

5. Conclusions

Static mixers are used in a wide range of industrial mixing challenges with the energy for mixing coming from the pressure drop across the mixer. The pressure drop in turn depends on the design, the fluid rheology, and the operating conditions. When a new product is introduced to a manufacturing site the design and operating conditions are largely fixed by the layout and ancillary equipment (eg pumps). The challenge lies in understanding the impact of the new product rheology on the capability of the mixer to achieve the desired outcome. Traditionally this would be done through empirical correlations and for static mixers this is the friction factor. If the correlations do not cover the parameter space, then additional experiments will extend the scope of the correlation. One such area is for viscoelastic rheologies where the published literature is sparse. A more effective way to expand the coverage would be through CFD simulations that incorporate linear and non-linear viscoelastic effects.

In this study we have examined the viscoelastic pressure drop enhancement using real process fluids representing high phase volume emulsions (shear thinning) and surfactant structured (viscoelastic) products, literature correlations and model fluids. A Giesekus constitutive model is found to provide a good description of the viscoelastic rheology. The CFD predictions of pressure losses for both inelastic and viscoelastic process fluids are in good agreement with experimental measurements. In addition, the friction factor coefficient for the mixer from both the experiment and the CFD are in good agreement with the value quoted by the vendor. However, for the viscoelastic fluid the literature correlation progressively over-estimates the friction factor as the Re increases by upto a factor of 4. On the other hand, the worst CFD predictions are less than a factor of 2 indicating that the CFD is a much better approach. Interestingly the experimental and CFD results for the viscoelastic fluid are parallel to the inelastic fluids but the correlation of Chandra and Kale using the original published constants diverges. A new set of constants for the correlation of Shah and Kale (1991) are obtained from the viscoelastic data as $R=0.267$ and $s=1.57$.

As a consequence of these promising results the impact of the flow field on laminar mixing of non-Newtonian fluids in

the SMX+TM has been discussed. The geometric arrangement of the crossbar elements induces regions of elongational flow that is favourable for dispersive mixing. These regions corresponded to peak values of the first normal stress differences that results in tension and compression of the fluid streamlines. Moreover, the impact of elasticity on the flow field is dramatic with stronger elongational flows for elastic fluids. In future studies we will expand on these findings to allow a more complete understanding of the mixing and processing conditions within the mixer, validated with experimental measurements.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix 1. – Validation of computational grid

Grid independence tests

Grid independence of the simulation results was ascertained using the value of the pressure drop across the mixer region (converted to friction factor) for a prescribed inlet velocity. Grid independence tests were performed using a short computational domain and established using relative error in the estimated friction factor values for a viscoelastic fluid (Giesekus model). A typical result is shown in Table 5 for simulations carried out at a mixer Reynolds number of 0.5 with a computational time-step of 10^{-5} s using 96 CPU cores (Intel Xeon E5-2690 v3 @ 2.60 GHz + 128GB RAM).

Newtonian benchmark

As a means of validation of the CFD simulations, the friction factor correlation for a Newtonian fluid (0.73Pas Glucose solution) was first computed. The results are compared against the empirical inelastic friction factor correlation for the mixer (Eq. 2) in Fig. 14 and show very good agreement. The results from the Newtonian fluid simulations were used to evaluate the friction factor ratio, $Z = f_{\text{mixer}}/f_{\text{pipe}} = 21.4$, which is in very good agreement with the value calculated from the pressure drop estimation supplied by the vendor, $Z = 19.75$ (Sulzer Chemtech Ltd, 2020) and the value calculated using Eq. 2, $Z = 20.2$.

Table 5 – Sample grid independence test results on DN50 mesh.

No. of cells	Friction factor	CPU time (h)	Relative error %
2510941	121.71	1.47	0.73
3455382	122.60	2.18	

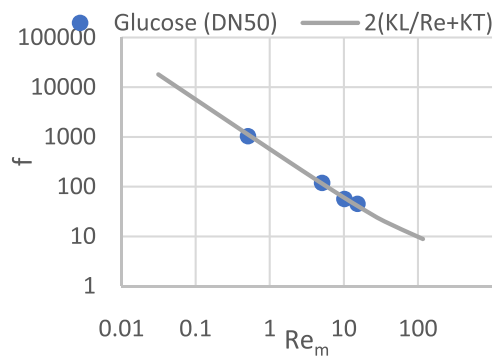


Fig. 14 – Friction factors for Glucose (Newtonian fluid). Symbols depict CFD results and solid line Eq. 2.

Appendix 2. – Rheology data

Oscillatory tests

Test fluid viscoelastic properties were measured using oscillatory techniques. The technique applied a stress or strain whose value was changing continuously according to a sine wave equation. The induced response (strain or stress) also followed a sine wave. The sample was continuously excited but never exceeded a certain strain and hence sample structure was not destroyed (providing steps were taken to keep the strain small enough).

Results for Mayonnaise are presented in Fig. 15. The phase angle remained roughly constant over the oscillation frequency range tested ($\delta = 6^\circ\text{--}9^\circ$), suggesting gel type properties.

Results for Shampoo are presented in Fig. 16. At low frequency (long timescales) $G'' > G'$ indicating shampoo is

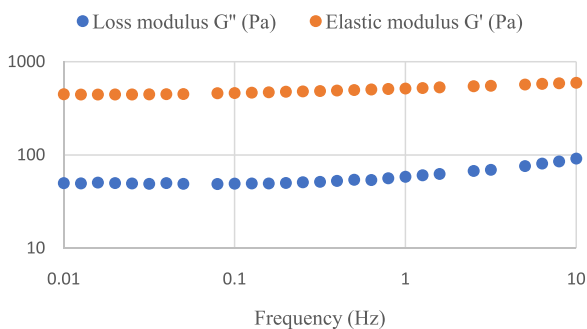


Fig. 15 – Frequency sweep results for Mayonnaise.

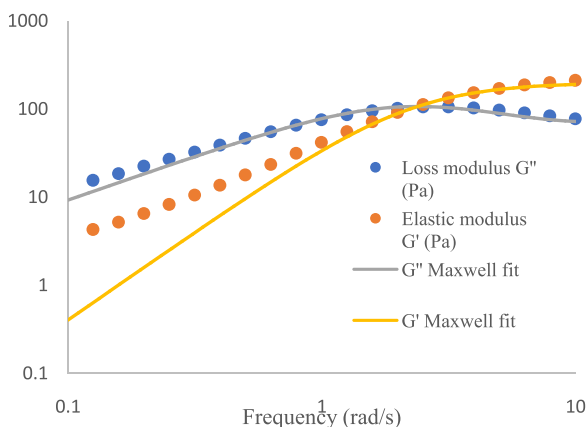


Fig. 16 – Frequency sweep results for Shampoo along with fit using Maxwell model.

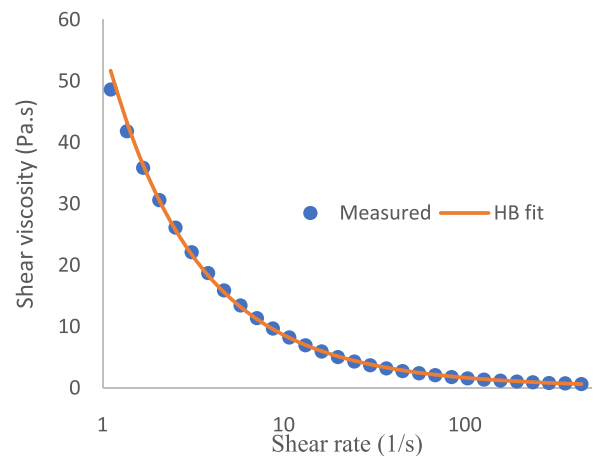


Fig. 17 – Viscosity curve for Mayonnaise obtained using ramped shear rate tests along with fit using HB model.

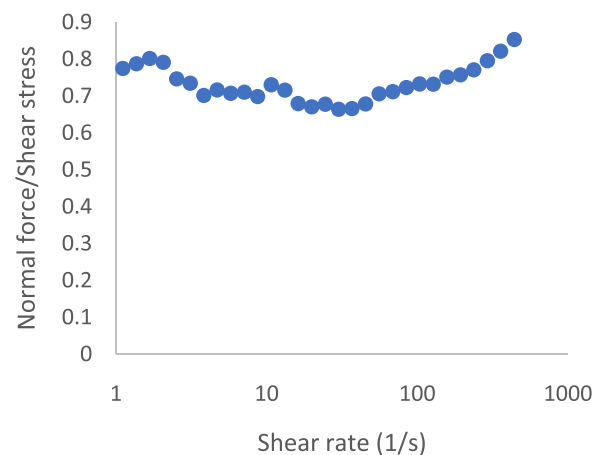


Fig. 18 – Data from the ramped shear rate tests on Mayonnaise presented as ratio of normal force to shear stress.

liquid-like whereas at higher frequency (short timescale) $G' > G''$ indicating solid-like behaviour. Phase angle increased with decreasing frequency, tending towards 90° at 0 Hz which was indicative of a viscoelastic liquid. A fit of the moduli data to the Maxwell equations (Eqs. 8,9) is also shown in the figure.

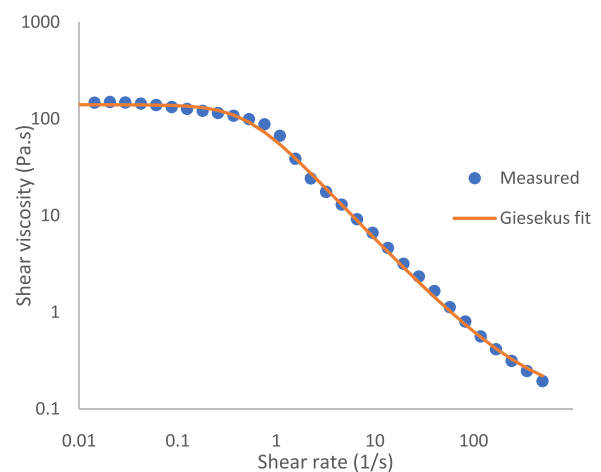


Fig. 19 – Viscosity curve for Shampoo using ramped shear rate tests along with fit using Giesekus model.

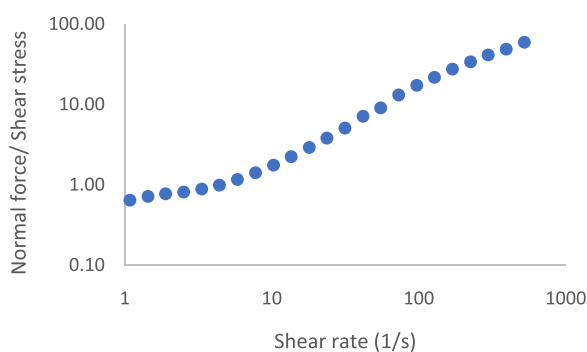


Fig. 20 – Data from the ramped shear rate tests on Shampoo presented as ratio of normal force to shear stress.

Flow curves

Basic flow characterisation was conducted using both stepped shear rate and ramped shear rate tests for each fluid sample. The shear rate ranges were chosen to span the pipe and mixer shear rates experienced by the fluids over the flowrate range tested during pressure measurements (1–600 1/s). A Herschel-Bulkley (HB) curve fit using the down ramp data was used for friction factor predictions as it best represented the shear history of the fluid entering the static mixer. The viscosity curve for Mayonnaise is presented in Fig. 17. The measured shear stress and normal force across the shear rate range are presented as a ratio in Fig. 18. This value is defined as W_{mixer} in the text (indicative of the strength of viscoelasticity) and can be seen to be < 1 across the range of shear rates measured confirming that Mayonnaise can be modelled as a viscoplastic liquid.

The viscosity curve for Shampoo using the data from the ramped shear rate test is shown in Fig. 19. The Giesekus equations (Eq. 10) are used to model this fluid and the fit to the data is also shown.

The measured shear stress and normal force across the shear rate range are presented as a ratio in Fig. 20. The fluid is strongly elastic particular in the higher range of shear rates that are representative of the mixer conditions (c.f. Table 4).

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